Digital Forensics Case Report: Web Application Preservation

CASE ID: 007 - Operation Web Mirror
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TARGET: https://momoboost.vercel.app

1. Executive Summary

A forensic website acquisition was conducted to capture and preserve a target web application for offline analysis. Utilizing HTTrack on an Ubuntu Linux environment, a complete mirrored clone of the client-facing website was successfully downloaded.

To ensure forensic integrity and defensibility, cryptographic hashing (SHA-256) was applied immediately upon acquisition and verified post-analysis. The cloned environment was then hosted locally via a Python HTTP server, allowing for safe, isolated examination of the site's layout, scripts, and static resources without tipping off the target server.

2. Scope & Environment

The objective was to create a reliable, point-in-time forensic snapshot of a potentially volatile web application, ensuring the evidence remained available even if the live site was altered or taken offline.

- **Host OS:** Ubuntu Linux (Root Access)
- **Acquisition Tool:** HTTrack Website Copier
- **Hosting/Analysis Tool:** Python3 HTTP Server
- **Integrity Algorithm:** SHA-256

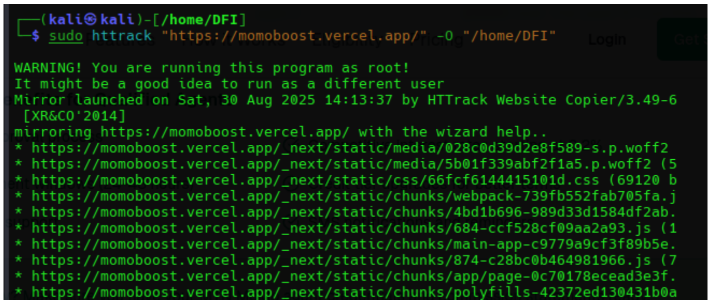
3. Acquisition & Chain of Custody

The acquisition followed a strict order of operations to ensure the evidence was captured safely and its integrity was mathematically proven.

1. **Site Cloning:** The target website was cloned using HTTrack with elevated privileges to ensure full directory write access.

```
sudo httrack "https://momoboost.vercel.app/” -O "/home/DFI"
```

HTTrack successfully downloaded the HTML, CSS, JavaScript, and linked structural resources.



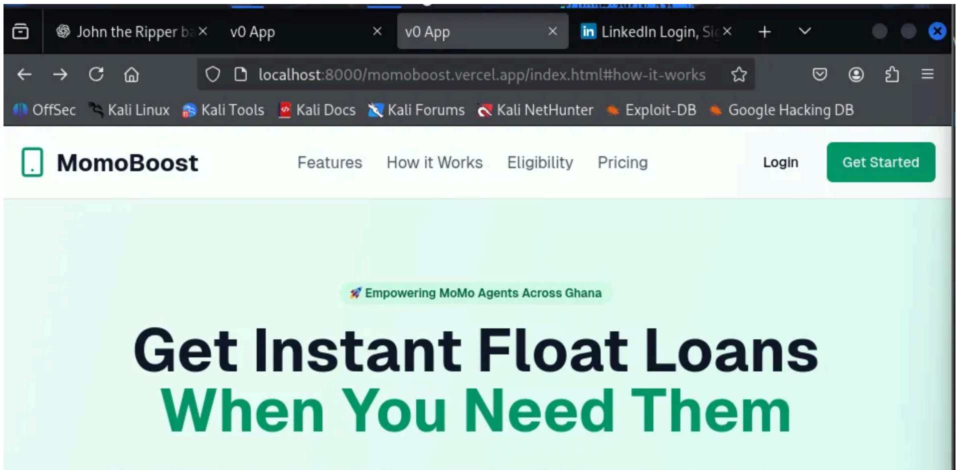
2. **Baseline Hashing:** Immediately after the download completed, a cryptographic hash of the entire directory was calculated and exported to establish a baseline for the Chain of Custody.

   ```
   sudo sha256sum -b /home/DFI/HTTrack_momoboost/* >
   ~/site_hash.txt
   ```

3. **Local Sandboxed Hosting:** To safely interact with the evidence without modifying the files, a local HTTP server was spun up within the cloned directory.

   ```
   python3 -m http.server
   ```

4. **Offline Access:** The mirrored site was successfully accessed via a web browser at: `localhost:8000`.
5. **Integrity Verification:** 12 hours post-analysis, the SHA-256 hash calculation was re-run against the directory. The hashes matched the baseline site_hash.txt perfectly, proving the evidence was not altered during the examination phase.



4. Technical Observations & Conclusion

The HTTrack cloning process successfully captured a highly accurate, static point-in-time snapshot of the target web application.

Key Observations:

- **Client-Side Preservation:** The cloned site was fully accessible offline. All page layouts, frontend scripts, and static media resources rendered correctly.
- **Server-Side Limitations:** As expected in client-side website forensics, dynamic content requiring backend processing (e.g., database queries or server-side logic) was not functional, as HTTrack only replicates client-facing source code.

Conclusion: This examination successfully demonstrated the methodology for volatile web evidence preservation. By isolating the web application offline and strictly enforcing cryptographic hashing before and after analysis, investigators are provided with a reliable, legally defensible snapshot of the digital evidence.